

FAQ

Q: Do I need to write my own optimisation algorithm, or can I use built-in functions?

A: You can use built-in functions, and there is no penalty for doing so. You can write your own optimisation algorithm if you want, but this is not expected.

Q: Does the power systems optimisation problem reflect real life?

A: Only to a limited extent – for example, maximizing cash after a year is not a realistic objective. However, some things may be applicable, e.g.: having diversity of supply if using intermittent generation may be a good idea.

Q: Bonus marks of 10% are offered for beating Jeremy's solution. Approximately what score is this?

A: The (sub-optimal) solution to which I will compare your solution scored a little below \$6.3 billion on last year's rng seed ('2025' in MATLAB). The seed will be changed again for 2026 (only revealed after submission so that solutions do not overfit a particular seed).

Q: I want more info passed to my functions. Is this okay?

A: Yes you can modify `evaluate_function()` to pass more inputs, but no information from a future time can be passed (e.g. you can have the mean and standard deviation of any random variable, but not the value it takes at a future time instant).

Q: I'm aiming for a top grade on this project. What are you looking for in an A+ project?

A: It is easy (almost trivial) to get a solution with a fairly good objective value, but much of the marking is on technical quality and depth. Here are some things to consider from a technical perspective (approximately from easier to harder; these aspects are not compulsory but serve as ideas):

1. Perform a comparison between two or more optimisation approaches. Explain your results.
2. After solving an optimisation problem, one of the usual aspects is to perform solution assessment, e.g. sensitivity analysis. Perform a sensitivity analysis for your solution, including:
 - a. sensitivity of the main design variables (or of any combination / transformation of the same);
 - b. sensitivity to important random variables (e.g. costs, capacity factors);
 - c. sensitivity to any other parameters which were not randomized but would have a significant effect on the solution.
3. At the start of each year-long simulation, the costs of each type of generation / storage are randomized (see details in code or on Learn). How should you adapt to this? E.g. if wind is very expensive one year, perhaps you don't want to build wind that year. But in other years, wind might be very cheap and part of the optimal solution.

How can the initial build plan be adapted to random costs, capacity factors, etc.? Develop / formulate a method for this, justify it (e.g. is it the solution to some fitting problem?), and show that it outperforms any non-adaptive build plan.

4. How should a battery be operated given changing prices, load and generation? This is a non-trivial problem in real-life, due to the difficulty of forecasting the future.

In this case, we have some idea of the future but the values are randomized so we don't know exactly. Can we use this information to operate the battery well?

5. Investigate whether any form of machine learning is applicable to this problem.
6. Again with reference to the operation of a battery: many optimal strategy problems over a time horizon are based on Bellman's principle of optimality (will be covered in my lectures, or use Google!).

Investigate whether the Bellman principle is applicable to this problem. Why or why not?